

Weijia Wang

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EDUCATION

University of Massachusetts Amherst (UMass Amherst)

Incoming M.S. in Geography; Ethical GeoAI / safe GeoAI

Amherst, MA

Starting Fall 2026

Northern Illinois University (NIU)

B.S. in Earth, Atmosphere and Environment-Geospatial Systems
2026

DeKalb, IL

Aug. 2024-May

GPA: 3.897/4.000; Summa Cum Laude; 2+2 Dual Degree Program with LJU

Awards: Harold A. & Marjorie K. Winters Scholarship Fund, NIU (2026); Richard E. Dahlberg Memorial Scholarship, Top 2%, NIU (2025); Dean's List, NIU (2025, 2024)

Lanzhou Jiaotong University (LJU)

B.S. in Geographic Information Science

Lanzhou, CHN

Sept. 2022-Jun. 2026

Award: Second Prize in 2023 Gansu Province College Student GIS Map Making Application Skills Competition, Top 10%, Provincial Level (2024)

PUBLICATIONS & PRESENTATIONS

Weijia Wang, Jichao Fang, Wei Luo. "Steering Instruction-Tuned Vision-Language Models for Multi-Label Landcover Classification." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), CV4EO Workshop, 2026. (Peer-reviewed, archival proceedings)

Weijia Wang, Jichao Fang, Wei Luo. "Parameter-Efficient Fine-Tuning of Vision-Language Models for High-Performance Earth Observation Tasks." AGU Fall Meeting 2025, New Orleans, LA. (Oral Presentation)

RESEARCH EXPERIENCE

Instruction-Tuned VLMs for Responsible Earth Observation

DeKalb, IL

Supervised by Prof. Wei Luo, NIU

Apr. 2025-Present

- Developed a parameter-efficient SFT and model-steering pipeline for instruction-tuned VLMs on Earth-observation imagery, framing multi-label land-cover tagging as a testbed for safe and responsible GeoAI evaluation
- Fine-tuned Qwen3-VL with 4-bit LoRA using RGB-only inputs and 1.7% class-balanced samples on BigEarthNet-S2, improving micro-F1 from 0.367 to 0.595 and macro-F1 from 0.253 to 0.523
- Compared prompt engineering, CoT zero-shot, and SFT baselines; analyzed failure modes and controllable precision-recall trade-offs through inference-time scaling
- Found adaptation concentrated in language-side MLP blocks and demonstrated LoRA sparsity: pruning 50% of LoRA weights retained micro-F1 0.562, while even 2% surpassed the zero-shot baseline
- Connected model adaptation results to AI-safety-relevant themes including robustness, interpretability, controllability, and responsible deployment of geospatial foundation models
- Resulted in a peer-reviewed WACVW/CV4EO 2026 paper and an AGU Fall Meeting 2025 oral presentation

Ground-Penetrating Radar (GPR) Inversion

Remote

Kaggle GprMax Deep Learning Challenge 1 (GDLC-1)

Oct. 2024-May 2025

- Designed a TransUNet inversion framework combining a CNN encoder, a 12-layer Vision Transformer, and a skip-connected decoder to reconstruct 2-D subsurface permittivity maps from multi-offset GPR B-scan data
- Preprocessed synthetic & gprMax-augmented data and implemented real-time physics-based data augmentation via gprMax FDTD simulations, tripling dataset size and improving generalization to unseen subsurface patterns
- Proposed a physics-aware composite loss function, consisting of MAE, SSIM, FFT, and gradient difference regularization in PyTorch to stabilize amplitude-phase inversion under electromagnetic constraints
- Achieved IoU = 0.873 on the hidden test set, ranked 1st among 14 finalist teams, and prepared the work for manuscript development in geophysical machine learning

Remote-Sensing LLM Prototype – VisualGLM-6B Fine-tuning & Deployment

Lanzhou, CHN

College Students' Innovative Entrepreneurial Training Plan Program, Instructed by Prof. Haowen Yan from LJU

Jan. 2024-Dec. 2024

- Built a domain-adapted VisualGLM-6B model to answer NL queries about Sentinel-2 imagery and Google Earth Engine assets, targeting applications in environmental change detection
- Fine-tuned VisualGLM-6B with LoRA (about 4M trainable params) on over 120k text-remote sensing pairs, and implemented GeoJSON-aware tokenizer for spatial AOI inputs
- Integrated a Vision Transformer encoder to process RGB+NIR previews, enabling multimodal prompt fusion and improving retrieval accuracy on AOI-based tasks
- Deployed the system using NVIDIA Triton Inference Server and Gradio dashboard, achieving < 2s latency per

- query on a single A100 GPU for interactive image search
- Delivered working prototype supporting text & map-based queries, and identified key failure modes like spectral aliasing and vague prompts

INTERNSHIPS

Henan Remote Sensing Institute

Zhengzhou, CHN

Intern in the Satellite Remote Sensing Application Center

Jun. 2025-Jul. 2025

- Processed and analyzed remote sensing imagery, including data preprocessing, feature extraction, and classification
- Addressed large-scale, repetitive tasks by experimenting with machine learning techniques to enhance workflow efficiency

Henan Academy of Agricultural Sciences

Zhengzhou, CHN

Intern at the Institute of Agricultural Information Technology

May 2025-Jun. 2025

- Collected, processed, and analyzed multi-source remote sensing data for winter wheat yield estimation, including visual interpretation of satellite imagery, UAV-based field surveys, and crop sampling
- Labeled deep learning training samples; explored machine learning & deep learning approaches to accelerate repetitive data processing tasks, improving efficiency under high data volume
- Conducted UAV flights and field sampling under extreme heat conditions, ensuring the timely completion of planned datasets

SKILLS

Computational Skills: Python, PyTorch, LoRA/QLoRA fine-tuning, Git, C++, SPSS

Professional Skills: ArcGIS Pro, ERDAS, ENVI, QGIS, Google Earth Engine, GeoDa